

AUTONOMOUS PLANT WATERING AND MONITORING ROBOT

MINI PROJECT REPORT

Subject Name and Code: MTE 2141 Sensors and Transducers

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MONTH & YEAR: November 2025

ABSTRACT

This project focuses on the development of an Autonomous Plant Watering and Monitoring Robot. It automates irrigation by combining sensors, ESP32 microcontroller, servo motor, and motor driver-based control. The robot autonomously navigates using IR sensors, monitors soil moisture levels, and waters plants using a relay-controlled pump only when required. The servo motor controls a soil probe arm that positions the sensor for accurate readings. This project promotes efficient water management and introduces practical mechatronics concepts applied in agriculture.

Keywords: ESP32, Smart Irrigation, Servo Motor, IR Sensors, Soil Moisture Sensor, Automation.

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

Agriculture and plant maintenance require efficient management of water resources. Over-irrigation and under-irrigation both result in reduced plant health and wasted resources. In modern-day scenarios, the use of automation and sensor-based systems in agriculture has gained immense importance. Automation reduces dependency on manual labor, saves water, and improves efficiency in small-scale and large-scale farming operations.

The motivation behind this project stems from the need to create a self-sufficient robot that can autonomously identify when a plant needs water and act accordingly. In urban gardening, research institutions, and smart greenhouses, such systems can greatly simplify maintenance and enhance productivity. By combining mechatronics, embedded systems, and control theory, this project aims to create a prototype robot that performs intelligent watering while navigating autonomously.

1.2 Present Scenario

Currently, irrigation is largely controlled either manually or through time-based systems. These approaches do not account for real-time soil conditions, leading to water inefficiency. Modern IoT-based systems have introduced smart irrigation methods that use sensors and cloud connectivity. However, these are often expensive or stationary. Mobile systems that can autonomously move between plants and perform watering based on soil moisture are still limited.

This project fills that gap by developing a low-cost autonomous robot capable of monitoring soil conditions and performing targeted watering. By leveraging the ESP32 microcontroller's computational capability and

PWM control, the robot can manage both navigation and irrigation simultaneously.

1.3 Project Scope

The project combines sensing, actuation, and control within a mobile platform. The soil moisture sensor provides real-time data for irrigation control, while two single-channel IR sensors assist in line-following navigation. The servo motor controls the soil probe arm that interacts with the ground for moisture measurement. The relay switches the pump ON or OFF based on sensor readings. All these modules are powered and coordinated by the ESP32 microcontroller. The scope extends to academic and research purposes, demonstrating the principle of autonomous mechatronic systems.

1.4 Project Work Schedule

Week	Task	Outcome
1-2	Component selection and circuit design	All parts finalized
3-4	Coding and basic sensor interfacing	Data verified from sensors
5-6	Integration of motor driver, relay, and servo	Robot assembled
7	Testing and calibration	Robot movement and control verified
8	Documentation and submission	Final report ready

1.5 Individual Objectives of Project Work

1. DEVANG RATHI (240929222)– Hardware interfacing and integration
- 2 SYED MOHAMMAD NAQVI (240929250)– Programming and sensor calibration
3. VASHISTH CHANDAK (240929256)– Mechanical structure and assembly
4. SAMUDRE ATHARV PRAVIN (240929306)– Testing, documentation, and presentation

Keywords: Automation, Irrigation, Embedded Systems, Team Work.

CHAPTER 2: THEORETICAL BACKGROUND

2.1 Present State and Recent Developments in the Work Area

Recent developments in smart irrigation include IoT and AI-based water control systems. These are often cloud-connected, allowing remote operation. The concept of mobile irrigation robots is an emerging field combining automation, robotics, and agriculture.

2.2 General Analysis

This chapter covers the working principles of the various components used in the project and their theoretical background.

The ESP32 microcontroller is a dual-core system-on-chip with integrated Wi-Fi and Bluetooth capabilities. It operates on 3.3 V logic and supports analog and digital interfacing. Its high-speed processing enables real-time decision-making for navigation and irrigation.

The soil moisture sensor operates based on the change in resistance as soil moisture varies. Wet soil conducts electricity better than dry soil. The sensor outputs an analog voltage proportional to moisture content, which is read by the ESP32's ADC pin.

IR sensors function using infrared light reflection principles. They emit IR light and detect the reflected beam; darker surfaces like black lines absorb IR, giving a LOW signal, whereas lighter surfaces reflect IR, giving a HIGH signal. This is crucial for line following.

The servo motor converts electrical PWM signals into mechanical angular movement, allowing precise control of the soil probe arm's position. The L298N motor driver allows bidirectional control of two DC motors, enabling forward, backward, left, and right movement. The relay module, on the other hand, acts as a switch for high-current loads such as the water pump, controlled via low-power ESP32 signals.

2.3 Tools Used and Their Details

- ESP32 – Main control unit
- IR Sensors – Detect black/white surface for navigation
- Soil Moisture Sensor – Measures dryness levels
- L298N Motor Driver – Controls the 2 DC motors
- Relay Module – Operates the water pump
- Servo Motor – Moves soil probe arm
- Buck Converter – Provides stable 5V supply
- Arduino IDE – For coding and debugging

Keywords: ESP32, Smart Control, Sensors, IoT, Automation.

CHAPTER 3: PROBLEM DEFINITION AND OBJECTIVES

The goal is to automate watering by monitoring soil moisture and enabling autonomous navigation. This addresses problems such as water wastage, manual intervention, and inconsistent irrigation.

Objectives:

- Design a robot capable of autonomous navigation
- Integrate soil moisture sensing and automated pump control
- Use servo motor for soil probe positioning
- Achieve reliable sensor communication with ESP32
- Build a compact, efficient, and cost-effective system

Keywords: Smart Farming, Problem Statement, Objectives, Embedded Design.

CHAPTER 4: METHODOLOGY

4.1 System Overview

The Autonomous Plant Watering and Monitoring Robot integrates sensing, control, and actuation. The ESP32 collects input from the IR sensors for navigation and soil moisture sensor for watering logic. Based on conditions, it sends output signals to the motor driver, relay, and servo. The robot moves along a line, stops near a plant, lowers the probe, checks moisture, waters if needed, and resumes motion.

4.2 Hardware Integration

The circuit connections are as follows:

- IR sensors (Left & Right) → GPIO 32 & 33
- Motor Driver IN1-IN4 → GPIO 26, 27, 14, 12
- Servo Motor → GPIO 13
- Relay (Pump Control) → GPIO 23
- Soil Moisture Sensor → GPIO 34 (Analog)
- Power Supply via Buck Converter (12V → 5V)

Common ground is maintained across all components.

4.3 Software Design

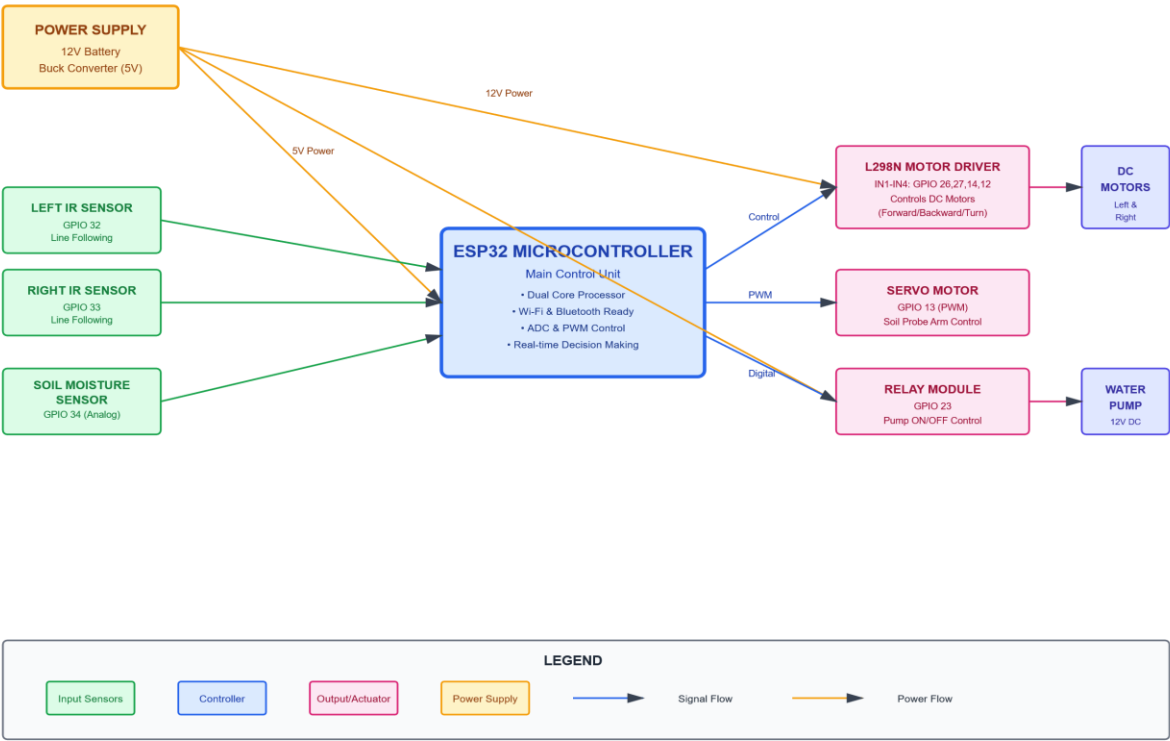
The software is implemented in Arduino IDE using the C++ language. The Servo library is used for servo control. The logic sequence is:

1. Move forward following the line using IR sensors.
2. Periodically stop and measure soil moisture.
3. If soil is dry, lower probe (servo), activate relay (pump ON), and water for a set time.
4. Raise probe and continue navigation.

PWM control adjusts motor speed, and delays are used for stable transitions. Testing confirmed the accuracy of sensor readings and the reliability of the relay during pump operation.

4.4 Block Diagram

AUTONOMOUS PLANT WATERING ROBOT - BLOCK DIAGRAM



CHAPTER 5: RESULTS, DISCUSSION, FUTURE SCOPE AND CONCLUSION

5.1 Results and Discussion

The robot successfully followed the line, measured soil moisture, and activated the pump based on dryness. Servo movement for probe positioning was precise and stable. The system demonstrated reliable performance during testing.

5.2 Future Scope and Limitations

Future developments include integrating Wi-Fi for data monitoring, solar power, and AI-based moisture prediction. Limitations include flat terrain operation and battery dependence.

5.3 Applications and Advantages

Applications

- 1. Smart Greenhouses:**

The robot can be used in controlled greenhouse environments to monitor and maintain optimal soil moisture for different plant species.

- 2. Urban Gardening and Home Automation:**

Ideal for small-scale indoor or rooftop gardens where manual watering is inconvenient.

- 3. Agricultural Research and Education:**

Useful as a demonstration model in academic labs to teach automation, robotics, and IoT concepts.

- 4. Precision Agriculture:**

Can be employed in farms to ensure each plant receives the right amount of water, reducing wastage and improving yield.

5. Nurseries and Plant Shops:

Helps maintain large numbers of plants efficiently, ensuring consistent care without constant supervision.

6. Remote or Unattended Locations:

Suitable for irrigation in areas that are difficult to access regularly, like remote farms or experimental fields.

Advantages

1. Efficient Water Management:

Waters plants only when required, preventing over-irrigation and saving water.

2. Automation and Reduced Manual Effort:

Fully autonomous operation minimizes human involvement in plant care.

3. Low-Cost and Scalable Design:

Built with easily available components, making it affordable and suitable for larger implementations.

4. Improved Plant Health:

Regular monitoring ensures plants receive adequate moisture, promoting better growth.

5. Energy Efficiency:

Uses minimal power, and can be upgraded to run on solar energy for sustainable operation.

6. Portable and Flexible:

The mobile design allows it to navigate between multiple plants instead of being fixed to one spot.

7. Educational Value:

Demonstrates the integration of electronics, sensors, and control systems, making it an excellent learning project for engineering students.

5.3 Conclusion

The project achieved all objectives, proving that low-cost embedded systems can automate irrigation effectively. It serves as a foundation for scalable smart farming solutions.

Keywords: Results, Future Scope, Smart Agriculture, Mechatronics Integration.

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